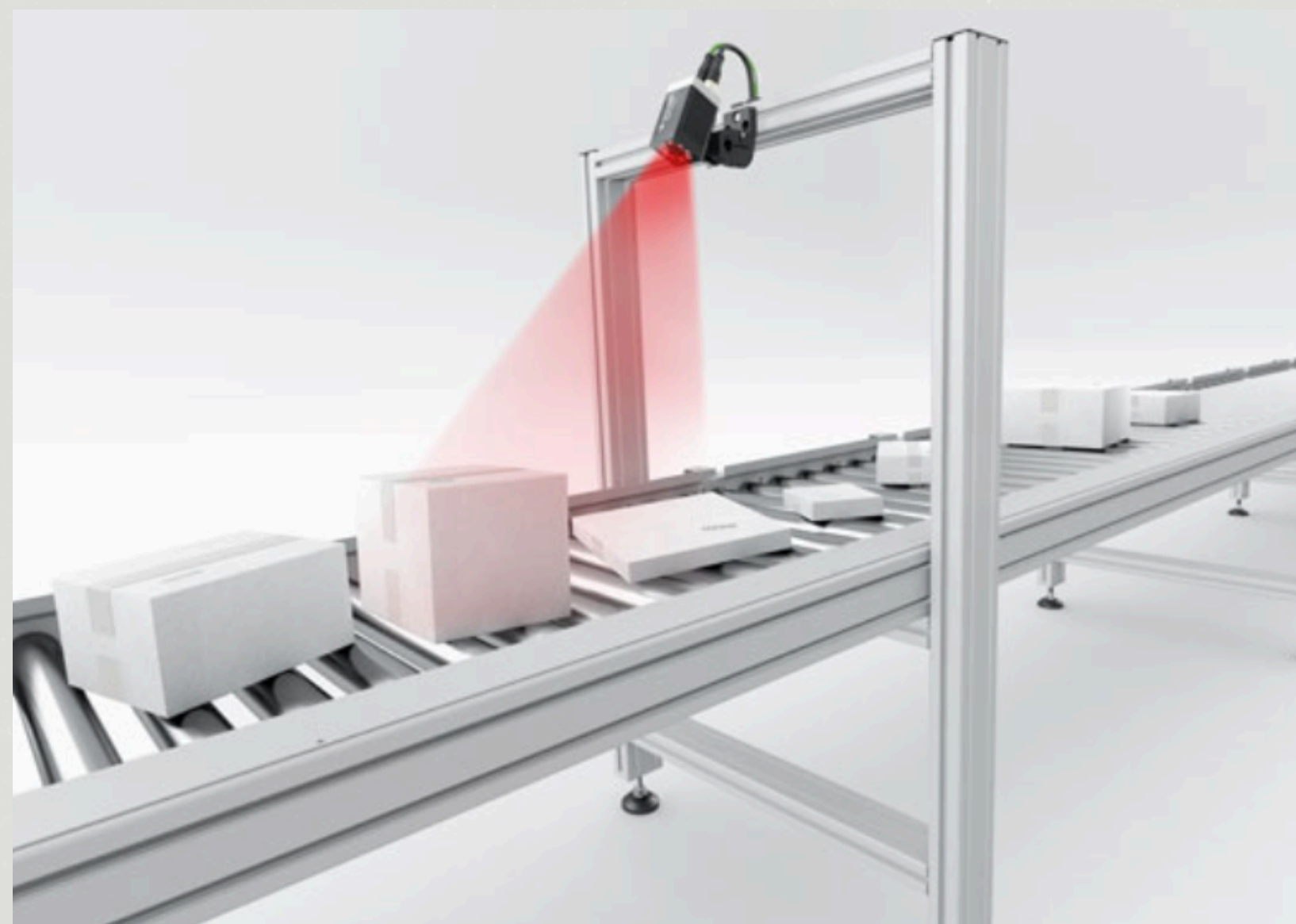


IsiDetector

Project Celio

Atanda Abdullahi



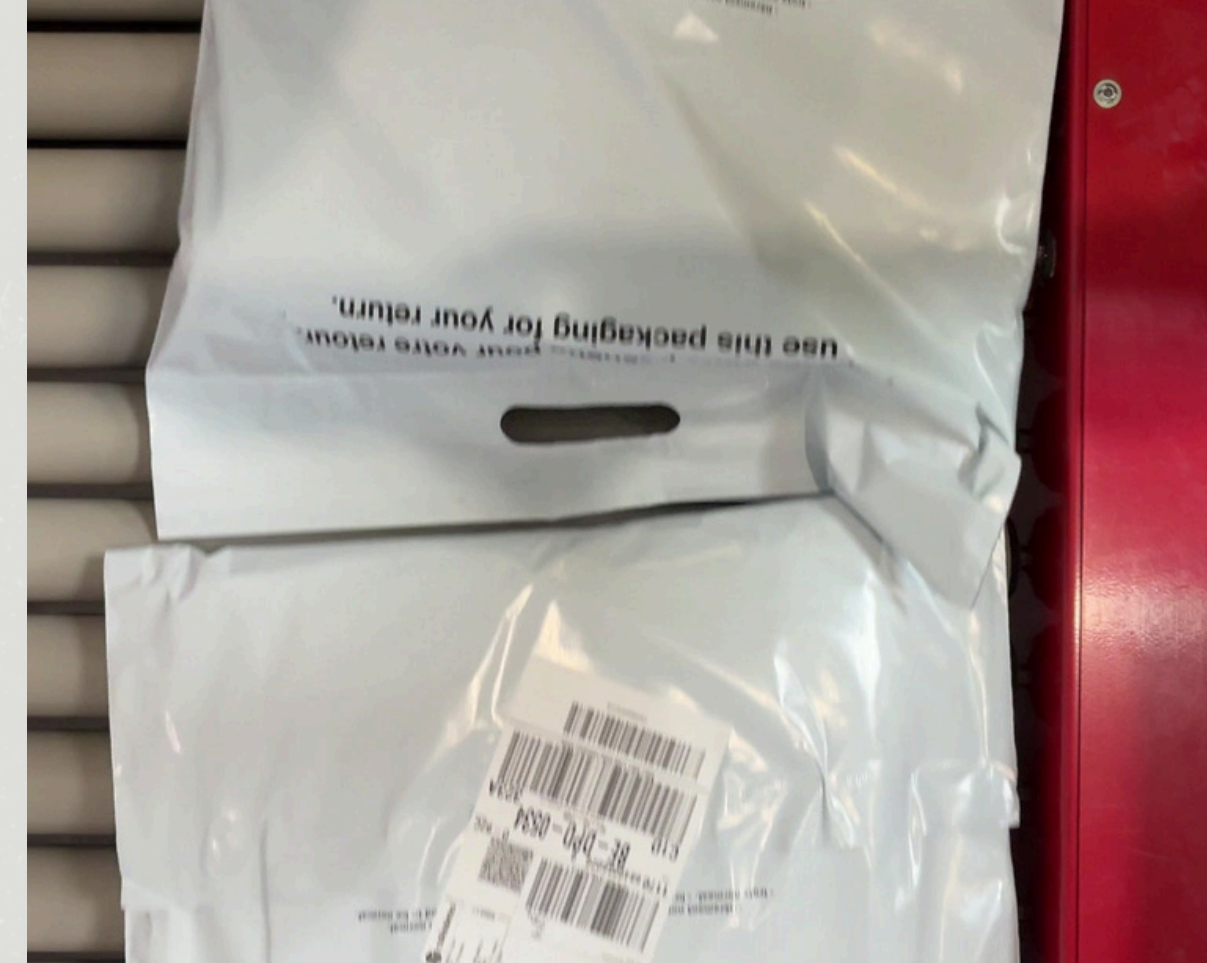
Industrial Computer Vision for Automated Parcel Sorting on Conveyor Belts.

- **Built as a product, not a script.**
- **YOLO + RF-DETR, GPU + CPU, Flask + FastAPI**
- **Linux + Windows, Office + Site**

Every time a parcel crosses, fires a UDP trigger to the sorter, fast enough that the gate has time to act, reliable enough that the belt never has to stop.

PROBLEM

- Cartons vs shiny polybags
- 1 m/s conveyor speed
- Glare, shadows, occlusion
- Precise triggering needed
- Runs on GPU & CPU
- Reliable , never stops



Objectives

ML

Can it detect?

isidet/ (preprocess, trainers, inference)

MLOps

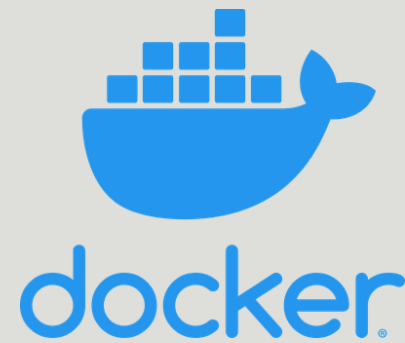
Can we ship, monitor, swap, audit ?

registry, hot-swap, compression, heartbeat, event log

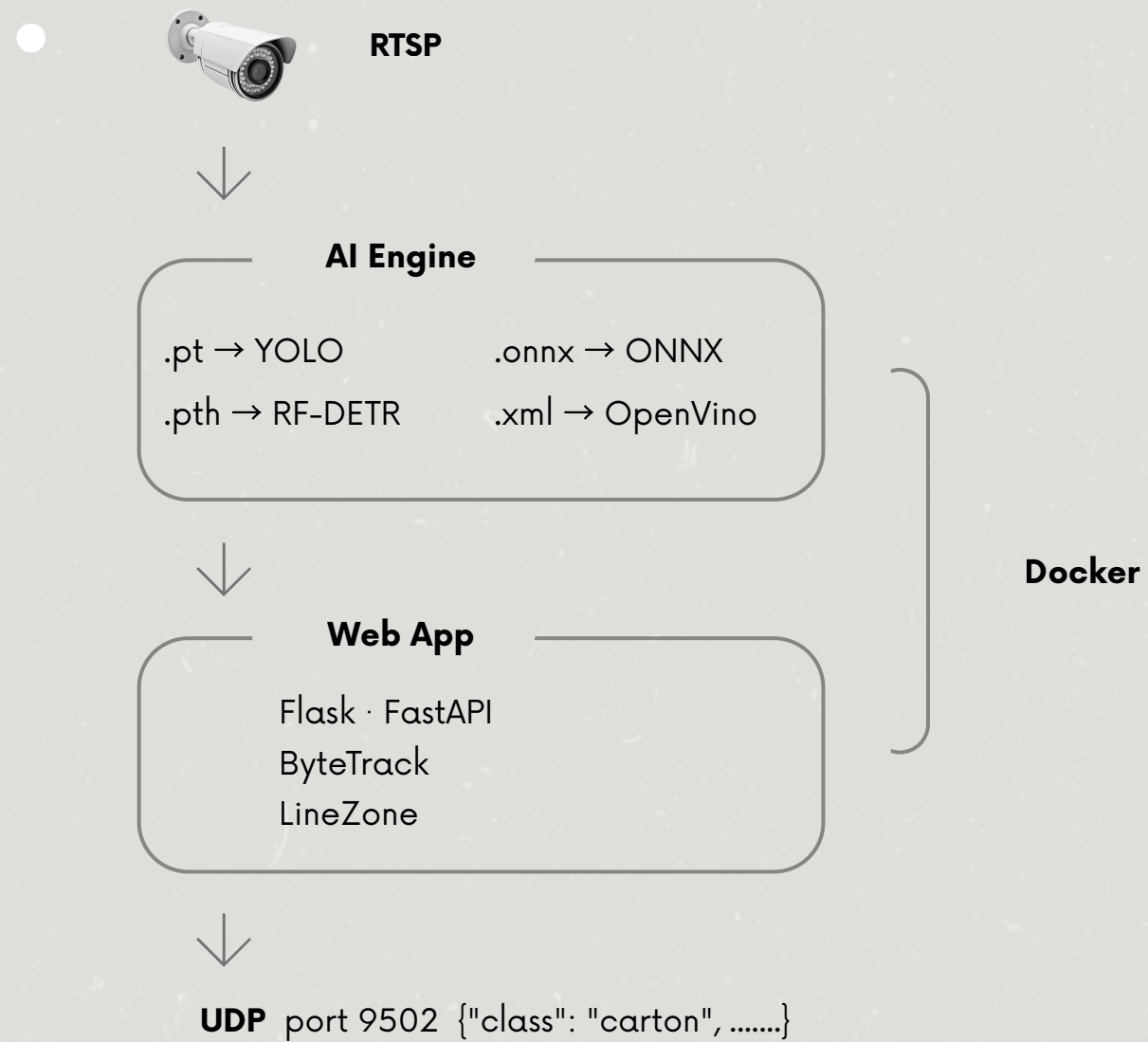
DevOps

Can we run it anywhere, on click?, non stop ?

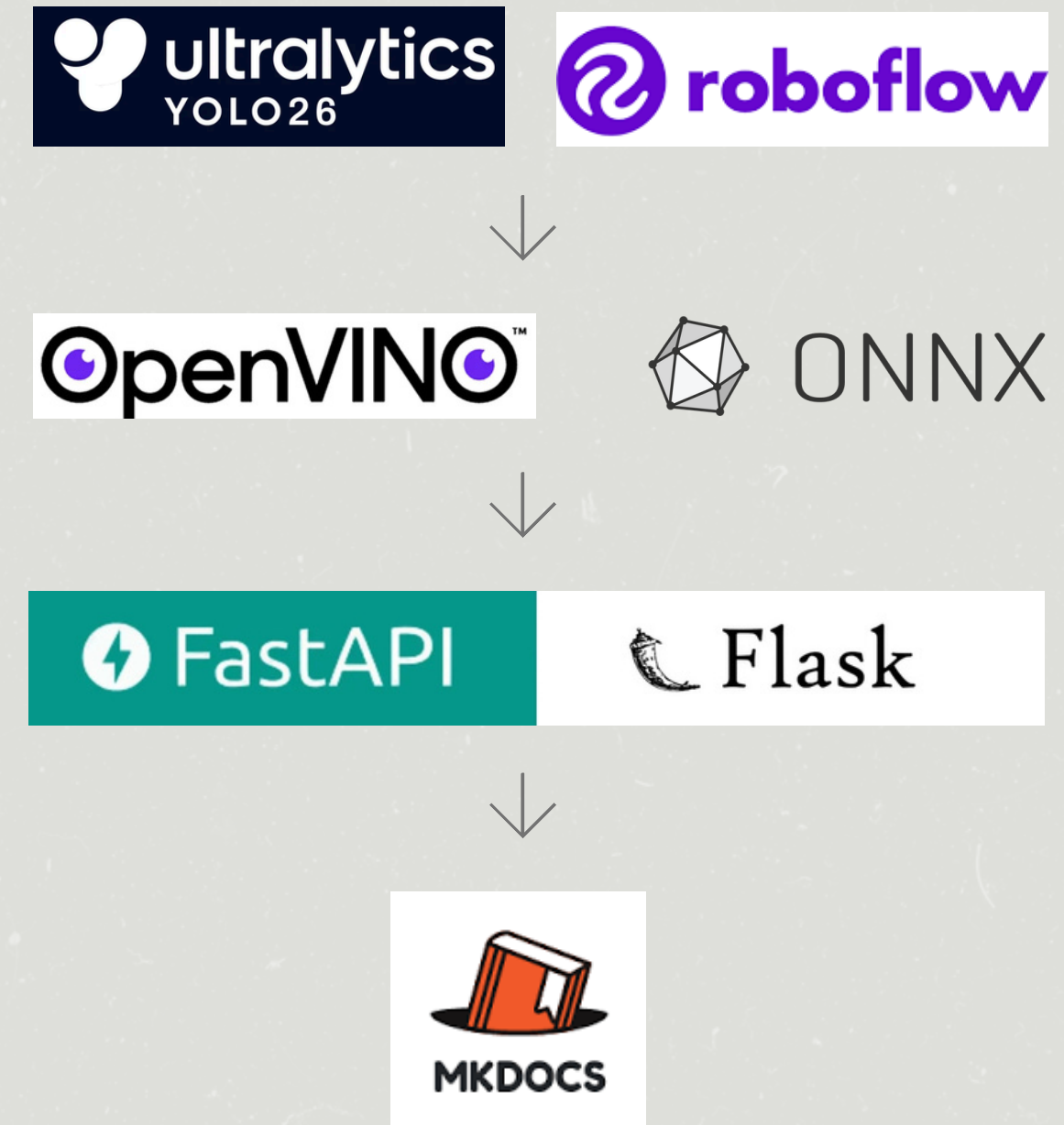
Docker, compose layering, kiosk, remote, branches



Architecture



- Dataset → Train → Compress → drop in models → engine auto-selects



key Features

Precision

Leading-Edge Trigger
ByteTrack (persistent IDs)
Reflection Filter

Reliability

Live Hot-Swap
Auto-reconnect (RTSP)
Heartbeat every 5 min

Observability

Event Log (CSV · 30 days)
Performance Dashboard
Analytics (24h · 7d · 30d)

Ops

Docker — ./up.sh
GPU + CPU
main (dev) · deploy (site)

Timeline

15/03/2026

Week 2

Week 3

Week 4

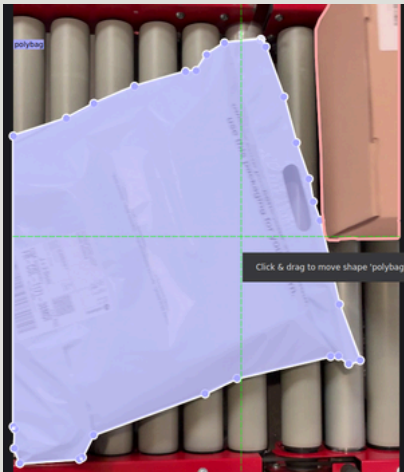
29/04/2026 Site Installation

Dataset Anotation
+ preparation

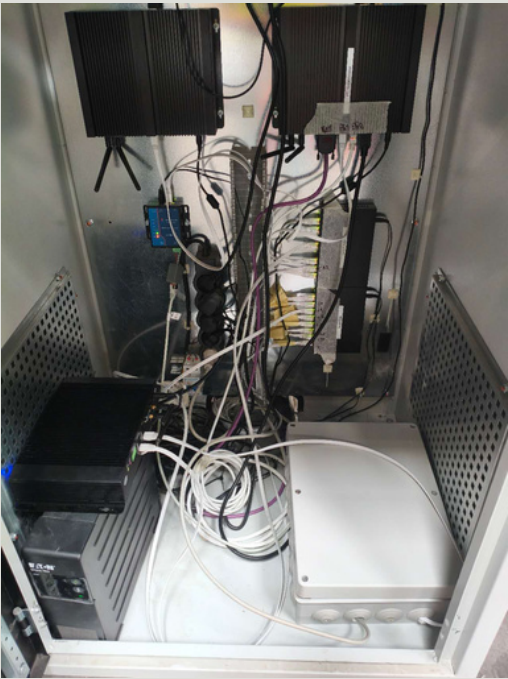
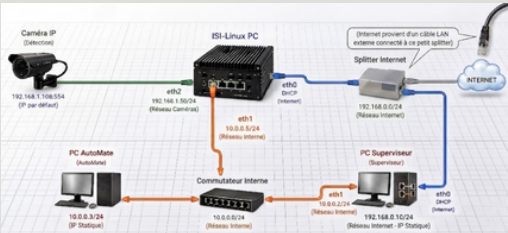
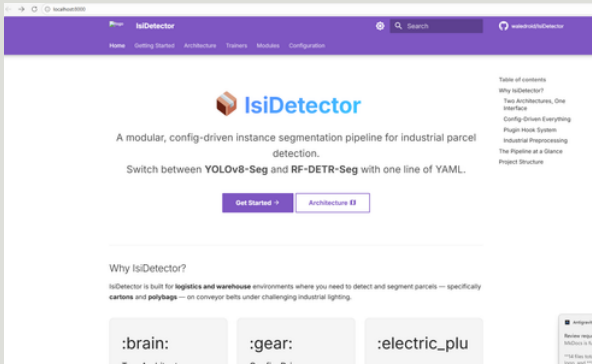
Reusable Trainer
+ Compression
+ Github

Inferencers
+ Web App
+ HotSwap
+ Performance
+ Docs

Docker
+ Deploy scripts
+ Test scripts
- camera
- network
- UDP Egress
- kiosk



2451/3141 images



Remote control script



Result

isitec

Plateforme ISITEC visionAI

🇬🇧

🇫🇷

Inférence en direct

Performance

Paramètres

Analytique

À propos

ANALYSE DE DÉTECTION

300

200

100

0

Mode 1

Source d'entrée

Caméra du site

Caméra enregistrée (configurable)

SimpleScreenRecorder

Recording

Pause recording

Schedule: (inactive)

Activate schedule

Edit schedule

☒ Enable recording hotkey

☐ Enable sound notifications

Hotkey:

☐ Ctrl +

☐ Shift +

☐ Alt +

☒ Super +

R

Information

Preview

Total time: 0:00:01

FPS in: 30.92

FPS out: 30.00

Size in: 1366x768

Size out: 1366x768

File name: simpl...0.mkv

File size: 4961 B

Bit rate: 0 bit/s

Preview frame rate: 10

Note: Previewing requires extra CPU time (especially at high frame rates).

Start preview

Log

tracyrecord::startInput() started input.

[PulseAudioInput::InputThread] Input thread started.

[FastResampler::Resample] Resample ratio is 1.0000 (was 0.0000).

Cancel recording

Save recording

BAGS

En direct

24h

7 Jours

30 Jours

Caméra

▶ DÉMARRER

■ ARRÊTER

📐 DÉFINIR ROI

🗑️ EFFACER ROI

UDP - CARTON #2284 @ 09:01:29

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(isi-train) aaatanda@ISIVISION:~/logistic\$./compress.sh

Isitec Model Compression Tool v1.0.0

Shrink your ONNX models for faster, smaller production inference.
Pick a strategy, feed it a model, ship a smaller artefact.

What this tool can do:

FP16 conversion

INT8 quantisation

TensorRT compile

Benchmark

Accuracy validate

Halve model size with ~no accuracy loss

Quarter size, ~1-3 % mAP drop, calibration-based

GPU-specific engine, 1.5-3x faster than ONNX-CUDA

Compare size + FPS across all variants of a model

Diff predictions against the baseline on a test frame

Detected ONNX Models

#	Family	Run	Files	Size	Variants	Source
1	yolo	23-03-20262	2	208.3 MB	base, sim	runs/segment/model...
2	yolo	yolo26m_416_200	2	179.8 MB	base, sim	runs/segment/model...
3	yolo	yolo26m_512_200	2	179.8 MB	base, sim	runs/segment/model...
4	yolo	yolo26m_640_200	2	208.3 MB	base, sim	runs/segment/model...
5	yolo	yolo26m_416_200	2	20.9 MB	base, sim	runs/segment/model...
6	rfdetr	10-04-2026_0110	2	246.7 MB	rfdetr-export, sim	models/rfdetr/10-0...
7	rfdetr	24-03-2026_0120	2	262.0 MB	rfdetr-export, sim	models/rfdetr/24-0...
8	rfdetr	25-03-2026_0043	2	262.0 MB	rfdetr-export, sim	models/rfdetr/25-0...

What would you like to do? [compress/benchmark/validate/refresh/exit] (compress):

DetectionObjet - 1.25.11.1

Plc Config

Efface

14_04_00_934 [UDP<- 192.168.2.49:36748] class=carton id=488 ts=2026-05-13T14:04:00.933250 (ClasseObjet=1)

14_03_57_700 [UDP<- 192.168.2.49:36748] class=carton id=487 ts=2026-05-13T14:03:57.690998 (ClasseObjet=1)

14_03_54_334 [UDP<- 192.168.2.49:36748] class=carton id=486 ts=2026-05-13T14:03:54.332497 (ClasseObjet=1)

14_03_50_973 [UDP<- 192.168.2.49:36748] class=carton id=485 ts=2026-05-13T14:03:50.972136 (ClasseObjet=1)

14_03_47_389 [UDP<- 192.168.2.49:36748] class=carton id=484 ts=2026-05-13T14:03:47.381274 (ClasseObjet=1)

14_03_43_689 [UDP<- 192.168.2.49:36748] class=carton id=483 ts=2026-05-13T14:03:43.691798 (ClasseObjet=1)

14_03_33_660 [UDP<- 192.168.2.49:36748] class=carton id=474 ts=2026-05-13T14:03:33.653231 (ClasseObjet=1)

14_03_30_944 [UDP<- 192.168.2.49:36748] class=carton id=475 ts=2026-05-13T14:03:30.931851 (ClasseObjet=1)

14_02_46_972 [UDP] Ecoute sur port 9502

14_02_46_957 [PLC] IsCom créé vers 192.168.2.45:1234

UDP Status:

UDP Online

What's next ?

CAMERA

GigE · 2–3 MP · 50 FPS,
Global shutter



LIGHTNING

Controlled lighting,
fewer false triggers



DATA

Retrain on site dataset,
accuracy tuned to Celio





Thank You